Contour-Preserving Flux Relabeling: A Reproducible Feasibility Study for Nonmagnetic Equilibrium Reconstruction

FUSIONZERO research working draft

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Preliminary research note. No original reconstruction breakthrough, cross-machine performance, or submission readiness is established.

Abstract

Equilibrium reconstruction from coil currents, plasma current, and electron Thomson measurements requires distinguishing contour geometry from the physical flux labels assigned to those contours. We investigate a monotonicity-constrained numerical reformulation of classical geometry-based reconstruction, rather than claiming a new physical identifiability principle. Removing the two Grad–Shafranov source functions on each contour produces a small quadratic program for the derivative of a scalar flux transformation. A current integral sets its scale, while an explicitly chosen reference sets its gauge. Independent analytic controls show second-order reduction of sampled-grid recovery error: for two shaped interior equilibria, relative errors decrease from 3.57×10^{-4} and 2.64×10^{-4} on 65^2 grids to 2.22×10^{-5} and 1.68×10^{-5} on 257^2 grids. Truncated contour coverage, measurement noise, incompatible geometry, and exterior matching expose distinct failure mechanisms. A discharge-separated DIII-D pilot establishes a local baseline score of 0.770653, below the challenge's cross-machine eligibility threshold. The relabeling intervention fails its source-validation physical gate: boundary derivative mismatch and raster-interface errors prevent acceptance. No MAST equilibrium labels are used, no blind evaluation occurs, and synthetic shape variation is not presented as experimental cross-machine transfer. The retained implementation and negative results define a reproducible starting point for a more consequential method.

1 Scientific question and evidence boundary

An axisymmetric tokamak equilibrium is often represented by the poloidal flux $\psi(R, Z)$, whose contours describe magnetic surfaces. The Fusion Equilibrium Challenge asks for this field and two scalar quantities without magnetic probe or flux-loop observations, using coil currents, total plasma current, Thomson electron profiles, and available geometry [1, 2]. Source training on DIII-D and zero-shot inference on MAST expose both diagnostic and geometric distribution shifts. Incomplete measurements do not imply that a unique hidden equilibrium exists.

This note asks a smaller falsifiable question: *if a predicted field has useful contour geometry but incorrect physical flux labels, can a scalar transformation correct those labels without moving the contours?* The mathematical reconstruction principle is classical. Bishop and Taylor give the chain-rule transformation and its exceptional degeneracy [3]; Christiansen and colleagues give a source-eliminating reconstruction and total current normalization [4]. The finite positive-spline formulation below is a numerical reformulation to be tested, not a claim of priority over those works. Regularized inverse equilibrium reconstruction is also established [5].

The closest inspected challenge entrant already uses an explicit coil/plasma decomposition, equal-temperature constraints, and physical diagnostics. It also reports testing a GS interior projection that lowered an equation residual while hurting its benchmark score; that projection

was rejected [6]. These author-reported experiments are not independently replicated here. Consequently neither "adding physics" nor lowering a residual establishes a contribution. Our admission conditions require accurate labels, identified numerical parameters, and compatible exterior matching. The current experiment does not satisfy all of them.

The dataset is pinned to revision `1e280905b85f2a6fdde7e06fca8cf3a1edf447cb`, and the official local scorer is version 3.2.0 [7]. The current diverted-only dataset differs from the original paper's counts. Only labelled DIII-D training discharges are used for local evaluation. MAST public inputs contain no equilibrium labels; older labelled demonstrations are explicitly excluded. No local MAST score can be inferred from an unlabelled input set.

2 Interior physics and conditional reconstruction

Use per-radian flux in Wb/rad, cylindrical coordinates in metres, and

$$\begin{aligned} B_R &= -\psi_Z/R, \qquad B_Z = \psi_R/R, \\ \Delta^*\psi &= -\mu_0 R^2 p'(\psi) - F(\psi)F'(\psi), \\ \Delta^* &= \partial_{RR} - R^{-1}\partial_R + \partial_{ZZ}. \end{aligned} \tag{1}$$

These equations assume static axisymmetry and scalar isotropic pressure. Electron pressure $p_e = en_eT_e$, with T_e in eV, does not supply the ion and fast-particle pressure. An additive flux constant affects neither the poloidal field nor Eq. (1); it must be fixed separately for absolute-label comparison. Dataset current signs must be converted consistently before a signed physical current claim is possible.

Let a smooth predicted shape field ϕ have connected regular closed contours $\Gamma_s = \{\phi = s\}$ on an annulus. We seek $\psi = h(\phi)$ with $h' > 0$. On each contour let P_s remove the span of 1 and R^2, under a declared positive sampling measure, and define

$$a_s = P_s\Delta^*\phi, \qquad b_s = P_s|\nabla\phi|^2. \tag{2}$$

Since the source functions in Eq. (1) are constant on a contour,

$$\Delta^* h(\phi) = h'\Delta^*\phi + h''|\nabla\phi|^2, \qquad h'a_s + h''b_s = 0. \tag{3}$$

Proposition 1 (Classical conditional labeling mechanism)**.** *Suppose $b_s \neq 0$, a_s is collinear with b_s, and $k(s) = -\langle a_s, b_s\rangle / \|b_s\|^2$ is continuous throughout the regular contour interval. Then all increasing labelings with an interior GS source representation have the form*

$$h(s) = D + C\int_{s_0}^{s} \exp\left(\int_{s_0}^{u} k(t)\,\mathrm{d}t\right)\mathrm{d}u, \qquad C > 0. \tag{4}$$

They differ only by affine output transformations. This is a conditional interior statement, not uniqueness among different contour geometries.

Proof. Dividing Eq. (3) by $h' > 0$ gives $h''/h' = k$; two integrations give Eq. (4). Conversely, Eq. (4) makes the projected operator vanish, so $-\Delta^*\psi = A(s)R^2 + B(s)$. Inverting h expresses A/μ_0 and B as $p'(\psi)$ and $FF'(\psi)$. Integrating them yields a formal source representation. Prescribed pressure, $F^2 > 0$, and physical boundary normalization require additional checks. □

The chain rule and exceptional family are not new results [3]. If $a_s = b_s = 0$, the surface supplies no information about h''/h'; if $b_s = 0$ but $a_s \neq 0$, no increasing relabeling can repair the source form. Finite sampling adds a separate rank failure even on nondegenerate contours. An optimizer can converge in all three situations. For example, $\phi = (R - R_0)^2 + Z^2$ has $b_s = 0$ on exact circles, but $\Delta^*\phi = 2 + 2R_0/R$ has a nonzero transverse component. No increasing exact repair exists. The finite QP can nevertheless report full rank, converge, and lower its sampled residual by pushing interior derivatives toward their lower bound. This independent-review counterexample is retained as a numerical control: the API's success flag means numerical solution and finite-basis rank only, never physical compatibility.

Contour preservation has a limited scope. Because $\nabla h(\phi) = h'\nabla\phi$, continuum critical points and contour sets coincide. At a critical point their Hessians differ by a positive scalar, so nondegenerate axis/saddle classification is preserved. However, a contour at normalized flux 0.95 generally changes its spatial location within this family. Neither q_{95} nor internal inductance is invariant. A masked interior transformation is not a global scalar transformation, and discrete interpolation or extraction can change the selected contour component.

3 Positive-spline numerical reformulation

Write $t = (\phi - s_0)/S$ on the modeled interval and choose nonnegative cubic B-splines B_j forming a partition of unity. Set

$$h'(\phi) = \sum_{j=1}^{m} c_j B_j(t), \qquad h''(\phi) = S^{-1}\sum_{j=1}^{m} c_j B_j'(t), \qquad c_j \geq \epsilon > 0. \tag{5}$$

The projected residual is linear in c, with columns $M_j = B_j a + S^{-1}B_j' b$. We solve

$$\min_{c\geq\epsilon} \ \|Mc\|^2 + \lambda\,\|Lc\|^2\,, \qquad \ell^{\mathsf{T}} c = I_\Omega, \tag{6}$$

where L is a second-difference matrix and

$$I_\Omega = -\mu_0^{-1}\int_\Omega \frac{\Delta^* h(\phi)}{R}\,\mathrm{d}R\,\mathrm{d}Z. \tag{7}$$

Equation (7) is a physical total plasma current only when Ω is the entire properly matched plasma region. In the analytic controls it is explicitly a bounded-domain source integral. Its equality prevents the trivial residual reduction obtained by collapsing field strength toward zero. The additive gauge is supplied independently.

We use $m = 12$, 64 contours, 128 arc-length samples per contour, $\epsilon = 10^{-5}$, and $\lambda = 10^{-10}$. Centering and scaling R^2 before QR projection avoids avoidable arithmetic conditioning. A cubic interpolant provides sampled-grid derivatives; exact analytic derivatives form an independent control. The normalization-augmented matrix rank is reported, as are surface support, positivity, and optimizer status. Full numerical rank is not a force-balance certificate.

For K contours and Q points per contour, projection and matrix assembly cost $O(KQm)$, a dense normal matrix costs $O(KQm^2)$, and its small dense solve costs $O(m^3)$ per active-set/Newton-type step. The implementation uses a generic constrained optimizer; no special convergence rate is claimed. Field interpolation, contour extraction, and the Nm grid-current assembly are additional costs. At fixed m, this is a small postprocessing problem, not a replacement for the missing free-boundary reconstruction.

3.1 Exterior matching is an independent gate

The identity $\Delta^*\psi/R = \nabla\cdot(\nabla\psi/R)$ gives, on a closed boundary $\phi = s_b$,

$$I_p = -\mu_0^{-1}\oint \frac{\partial_n\psi}{R}\,\mathrm{d}\ell = h'(s_b)I_\phi. \tag{8}$$

Keeping an already matched exterior requires both the boundary flux and normal derivative to agree. In general, a scalar relabeling cannot repair arbitrary current error while leaving that exterior unchanged. Matching only the flux value leaves a poloidal-field jump; a pasted raster intervention must be measured again rather than relying on its interior chain-rule residual.

There is also a global obstruction. If $\Delta^*\phi = 0$ and $|\nabla\phi| > 0$ in vacuum, then preserving vacuum under a global h forces $h'' = 0$ on the flux values attained there. If those values span the interior interval, the transformation must be affine. A global synthetic corruption of both interior and vacuum is a different problem from preserving an already valid physical exterior. The present controls cover only the interior source form.

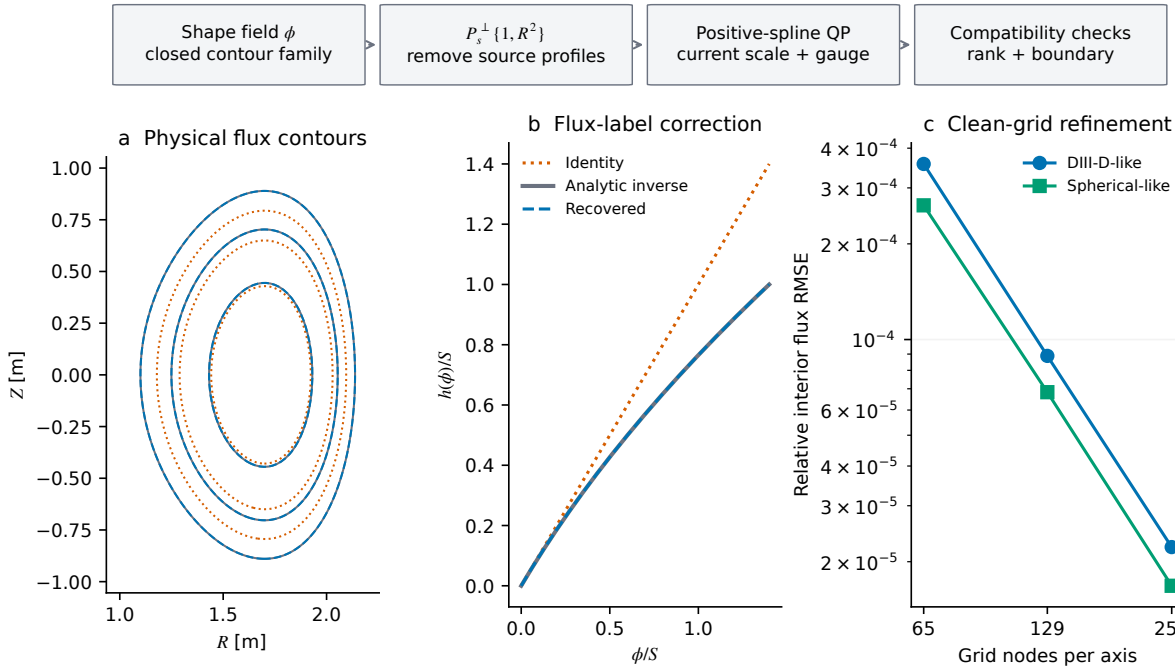


Figure 1: Classical flux-label reconstruction, reformulated with positive spline coefficients. (a) Three fixed physical-flux levels: the true contours are gray, the nonlinear corruption is dotted orange, and the recovered labels are dashed blue. The curves remain in the same family, while the contour carrying a particular physical flux value changes. Coordinates and aspect ratio are physical. (b) Recovered and analytic inverse labeling nearly coincide on the clean control. (c) Sampled-grid errors decrease with refinement in two analytic interior shapes. No vacuum-matched boundary or experimental machine transfer is demonstrated. All panels are regenerated from retained arrays.

4 Reproducible experimental design

4.1 Independent analytic controls

The independently authored generator uses

$$\psi_0 = K\left[\frac{(R^2 - R_0^2)^2}{8R_0^2} + \frac{Z^2}{2\kappa^2}\right], \qquad \Delta^*\psi_0 = K\left(\frac{R^2}{R_0^2} + \kappa^{-2}\right). \tag{9}$$

With a declared flux scale S, set $x = \psi_0/S$ and $\phi = S(x + 0.4x^2)$. This has a known increasing inverse. Two parameter sets vary aspect ratio and elongation; their names "DIII-D-like" and "spherical-like" denote analytic shapes, not models validated against either machine. Neither has a verified diverted boundary or vacuum-matched exterior. Errors are evaluated on $0.1 \le x \le 0.9$, with the declared current quadrature over $x \le 1$. The discrete source integral, not a silently substituted continuum integral, normalizes the discrete solver.

Controls include identity and affine labels, a contour-moving perturbation, smooth perturbations, independent sample noise, and degenerate geometries. Fixed noise amplitude is not expected to converge under differentiation: finer resolution can increase derivative noise. A first run with contours only between $x = 0.15$ and 0.85 is retained as a failed experiment; two spline columns were unobserved. The revised sampling covers $x = 0.01$ to 0.99 and explicitly checks augmented rank. This revision preceded any use of real-data relabeling results.

Table 1: Sampled-grid recovery of a known nonlinear flux-label corruption. Relative RMSE is measured on a fixed analytic annulus, with the synthetic gauge declared in advance. Shape variation is not experimental machine transfer.

Interior shape	65^2	129^2	257^2
DIII-D-like	3.57×10^{-4}	8.88×10^{-5}	2.22×10^{-5}
Spherical-like	2.64×10^{-4}	6.83×10^{-5}	1.68×10^{-5}

4.2 Discharge-separated DIII-D pilot

We select 128 labelled training discharges by sorting a seeded SHA256 function of their public relative filenames. Indices modulo eight allocate 96 training, 16 validation, and 16 test discharges. Each contributes 16 evenly spaced available equilibrium timestamps, yielding 1536/256/256 frames. Entire discharges remain disjoint. This is a small pilot, not a full benchmark result.

Inputs are the permitted currents and offline-aligned core/edge Thomson features. A documented source erratum corrects the plasma-current clock. Missingness indicators and all imputation, scaling, target standardization, and 32-component PCA are fitted using training frames only. Seven fixed models compare a mean map, two ridge models, two radial-basis kernel ridge models, and two extra-tree regressors. Selection minimizes a validation latent-plus-scalar loss fixed before evaluation. All predetermined candidates are reported; test scores do not select the model. The baseline's raw channel summaries are source-specific and are not a machine-independent representation.

The local composite score is

$$\begin{aligned} S = {} & 0.55 \max(0, R^2_\psi) + 0.15 \max(0, R^2_{q,\beta}) \\ & + 0.10\big[1 - \min(1, D_{\mathrm{LCFS}})\big] + 0.20C. \end{aligned} \tag{10}$$

Here $R^2_{q,\beta}$ is the mean of the two pooled scalar R^2 values, and C is the mean of the individually clipped $\max(0, R^2_j)$ values for the seven map-derived quantities having nonzero reference variation. The official derived-geometry term C is not the GS residual or a measured current-closure check. The pinned reference envelope enters evaluation only. Prediction correction uses its own contours without reference-derived masks. Mean shot flux RMSE intervals resample entire test discharges 2000 times; they are conditional small-cohort intervals, not official-score intervals.

5 Measured results and failed admission

The clean controls pass the preregistered 10^{-3} recovery threshold and show approximately second-order improvement under refinement (Table 1). At 129^2, nonlinear input-label errors of roughly 0.28 become below 9×10^{-5}. An affine current rescaling alone does not undo a nonlinear distortion. This validates the implemented controlled inverse, not a novel principle or a reconstruction from incomplete diagnostics.

The initial truncated-coverage run achieved a small projected residual while retaining relative label error around 0.26. Its successful optimizer return did not identify its null directions. The retained full-rank controls also show that a source-design matrix can have rank two while its transverse gradient component vanishes. Checking only the source-design rank is insufficient. Noise and contour-moving perturbations leave residuals and recovery errors that do not vanish through the clean-control mechanism.

The validation-selected pilot model is radial-basis kernel ridge. On 16 held-out test discharges its local S is 0.770653, $R^2_\psi = 0.885775$, $R^2_{q,\beta} = 0.673734$, $D_{\mathrm{LCFS}} = 0.059525$, and $C = 0.441846$. Mean shot flux RMSE is 0.052302 Wb/rad, with a conditional 95% whole-shot interval $[0.034941, 0.072872]$ Wb/rad. The 0.85 source-score eligibility threshold is not reached.

Table 2: All fixed DIII-D pilot baselines on 16 test discharges (256 frames). Selection uses the predeclared validation loss, not test scores. These are small-cohort local measurements, not an official leaderboard comparison.

Model	S	R^2_ψ	D_{LCFS}	C
Mean	0.245538	0.283129	0.10183	0.00000
Ridge, $\alpha = 1$	0.599889	0.700657	0.05889	0.29301
Ridge, $\alpha = 100$	0.620455	0.721526	0.06476	0.30346
RBF, $\gamma = 0.1/d$	0.702212	0.785260	0.05726	0.43583
RBF, $\gamma = 1/d$ (selected)	0.770653	0.885775	0.05953	0.44185
Extra trees, leaf 1	0.666394	0.720095	0.06508	0.38272
Extra trees, leaf 4	0.642205	0.695857	0.06815	0.34121

A 256-frame CPU prediction batch took 0.010886 s median, excluding raw-input preprocessing and scoring; this does not establish real-time reconstruction. Table 2 reports every fixed comparator. Representative and failure maps are shown in Fig. 2; absolute flux error includes gauge and scale error and cannot alone establish a physical improvement.

The raw relabeling intervention is tested only on the separate 256-frame validation set. Every frame has a boundary-derivative mismatch above 10^{-6}. Its median relative edge mismatch is approximately 0.0343, and the implied median poloidal-field jump is approximately 0.00848 T. The median source-form residual changes from approximately 0.0193 to 0.0177 in the interior analytic fit, but rises to approximately 0.0387 after pasting the transformation into the actual raster map. These are three different numerical objects and must not be conflated. The physical admission gate fails; no corrections are applied to test frames. Exact final tables and retained map hashes accompany the experiment report.

The completed validation scores are 0.860378 (unchanged), 0.839649 (affine), and 0.839617 (nonlinear). The nonlinear change is -0.020761, failing the preregistered 0.005 loss bound; mean shot RMSE increases 1.92%. These are reused baseline-selection validation discharges, not an independent confirmation cohort. Baseline test performance was already reported; the intervention was not selected or tuned with test labels. The median contour change is small, but the largest extracted-contour change is 2.5671 m, illustrating possible component-selection failure after a masked transformation. The maximum relative pasted-current error is 64.39; that frame has a permitted input current of only 2996 A and a much larger predicted current-region integral. Clock/support validity is unresolved, so this extreme must not be attributed solely to the relabeling mathematics. The numerical boundary check also samples a contour with a small spread of flux values: a future acceptance gate must check value and derivative pointwise, rather than only h' at a representative boundary value. This limitation cannot rescue the current result, since every frame already fails.

6 Interpretation and remaining research

The controlled inverse separates a tractable label error from geometry error, but the source-validation experiment blocks a useful full-equilibrium claim. Preserved interior contours do not supply compatible external fields, correct profiles, or consistent scalar predictions. Measured-current normalization must respect the boundary identity, and an unqualified interior repair can introduce the very physical defect it was intended to remove.

The appropriate next experiment is an independently matched exterior-aware formulation, or a method that obtains additional geometric information from permitted diagnostics before correcting the labels. A separate, preregistered measurement experiment tested joint electron-temperature/density profile registration. An input-only audit found only 14 of 2048 sampled DIII-D core profiles with a smoothed temperature maximum at least four channels from each

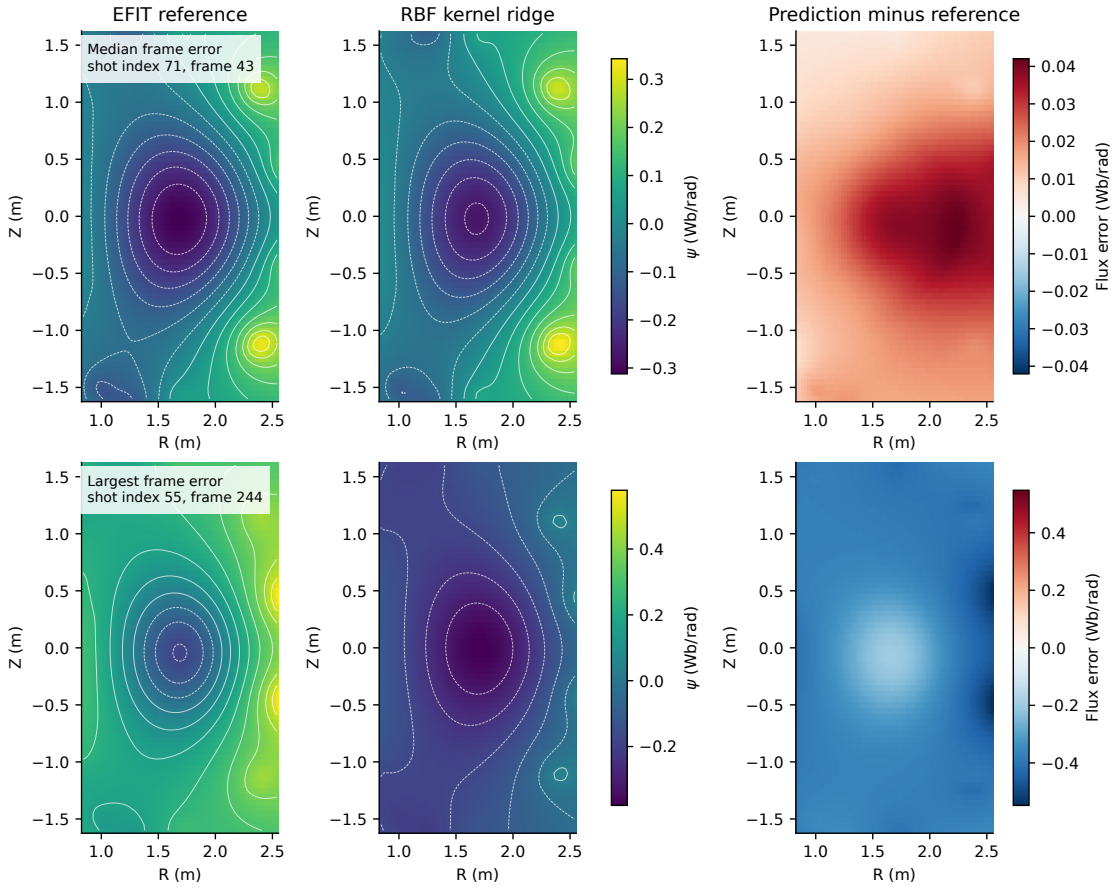


Figure 2: Real held-out DIII-D examples from the validation-selected baseline: the test frame at median flux error (top) and the largest frame error (bottom). Reference and prediction share a color scale within each row, and difference maps use symmetric error scales in Wb/rad. Physical geometry is preserved. Examples are selected by a declared error rule, not as a population estimate. The large failure contains substantial offset/amplitude error; an additive gauge correction alone would not constitute a physical advance.

end; this peak test does not locate the magnetic axis or prove flux coverage. Consequently the experiment matched partial core and tangential-chord profiles, with training-only joint feature scaling, timing and separation guards, orientation comparison, and an abstention option.

The tested classical one-to-one local sequence alignment also failed its improvement gate. Its frozen test configuration accepts 82 of 184 input-eligible frames, producing 620 pairs across 15 discharges. On exactly those core queries, mean discharge-averaged absolute flux mismatch, normalized by the reference axis-to-boundary span, is 0.092475 for ordered joint matching, 0.067644 for equal-temperature nearest matching, 0.054406 after isotonic temperature smoothing, and 0.068248 for joint nearest matching. Reference flux enters evaluation only. These results reject this particular one-to-one formulation; unequal channel sampling, calibration, and partial overlap leave continuous and many-to-many formulations as different untested questions. A separately labelled evaluator audit corrects a secondary coverage statistic that was conditional on ordered-matcher acceptance; it does not change these query-matched comparisons. No candidate is promoted to a full equilibrium solver or claimed to generalize experimentally to MAST.

Originality, meaningful superiority over strong comparable reconstructions, and zero-shot experimental MAST performance remain open. A high score alone would not resolve the missing physics, and the present low score does not prove that the diagnostic information is absent. Control deployment and independent reproduction remain unverified.

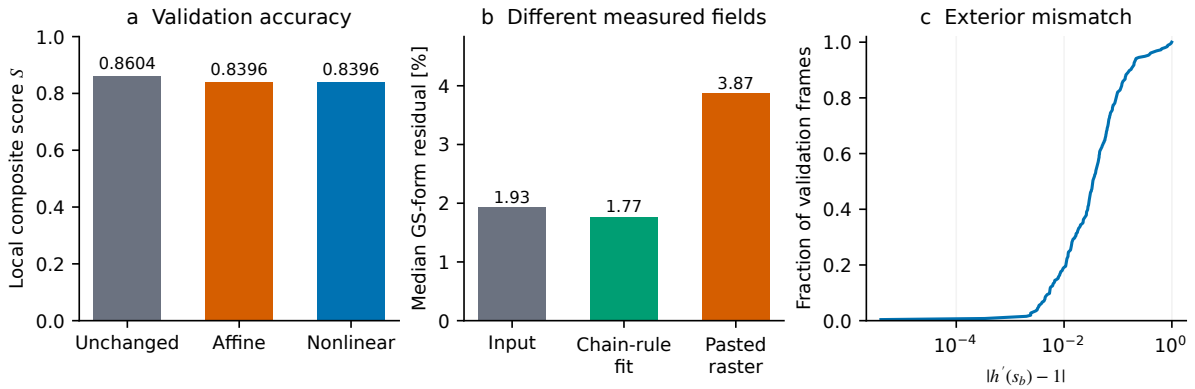


Figure 3: Validation-only falsification on 256 frames from 16 discharges. (a) Both interventions reduce the local composite score. (b) The chain-rule fit has a slightly smaller median source-form residual, but the actual pasted raster has a larger one; these are different fields. (c) Every frame has an exterior derivative mismatch. The intervention is rejected and no test-map corrections are made. Curves show descriptive frame distributions, not independent-sample confidence intervals.

Reproducibility and status. The project retains pinned public source manifests, raw-data URLs and hashes, discharge splits, compact processed arrays, fixed protocols, executed source snapshots, all baseline predictions, analytic fixtures, and failed runs. The numerical/data test suite checks inverse recovery, refinement, rank abstention, current-sign incompatibility, clock handling, label isolation, discharge separation, and training-only PCA. Scorer self-checks return one for perfect predictions and zero for zero predictions. CPU hardware, dependency versions, timings and measurement exclusions are recorded. All results are local; there is no official hidden-label evaluation. This working draft is not ready for submission as an original reconstruction breakthrough.

Data acknowledgement. The challenge dataset is released by Sophelio and collaborators under CC BY 4.0. Its DIII-D data acknowledgement credits the U.S. Department of Energy, Office of Science, Office of Fusion Energy Sciences, and the DIII-D National Fusion Facility under Award DE-FC02-04ER54698 and the additional awards listed in the pinned dataset card. This credits the dataset's support; it does not assert that the present study was funded or endorsed by those organizations.

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